

AIGC Inference Trace Integration Architecture (M4)

① Trace Sources

uniform preset

prompt $\in [64, 1024]$ tokens
output $\in [50, 500]$ tokens
uniform random sampling

lognormal preset

prompt $\sim \text{LogN}(\mu=5.5, \sigma=1.0)$
output $\sim \text{LogN}(\mu=4.5, \sigma=1.2)$
long-tail distribution (Azure trace)

ModelCatalog

llama-7b / llama-13b
llama-70b (INT8) / sdxl
Zipf($\alpha=1.2$) model selection

② Workload Generator (environment/task.py)

Task.generate_inference_workload(N, rng, dist, arrival_rate)

for each request i in $[0, N)$:

- sample `model_id` from catalog
- sample `prompt_tokens`, `output_tokens` from `dist`
- `arrival_time[i] = arrival_time[i-1] + Exp(arrival_rate)`
- produce (Prefill, Decode) pair sharing `req_id`

N × (Prefill → Decode) DAG

each task carries: `model_id`, `kind`, `prompt/output_tokens`, `arrival_time`, `kv_cache_GB`

③ Simulation Environment

Tasks (Simulation.step)

WAITING → READY:
✓ `deps` satisfied
✓ `current_time` $\geq$ `arrival_time`

READY tasks

Scheduler

baselines: RR / HEFT / GA / PSO
ours: AIGC-aware PPO
(state: `kind/model/loaded/batch/affinity`)

dispatch

Servers (1 Cloud + N Edge, heterogeneous)

Cold load + LRU eviction
(M1)

KV cache in VRAM
(M2 step 3)

Continuous batching
(M3 step 1)

Memory-bandwidth floor
(M4 step 2)

④ Metrics Collection (per checkpoint)

Common metrics (all workloads)

`makespan` · `avg_e2e_latency` · `avg_utilization` · `load_balance_std`

AIGC metrics (inference workload only, M4 step 1)

TTFT P50/P95/P99 · TPOT P50/P95 · goodput · SLO attainment · req_e2e P50/P95

⑤ Output Files (under --out /DIR/)

`benchmark_raw.csv`
per-run, per-checkpoint

`benchmark_summary.csv`
mean / std / 95% CI

`statistical_tests.csv`
Mann-Whitney U test

`run_manifest.json`
config + git commit